

Appendix to: Modelling of a Galactic Flyby and of Stellar Formation On Stellar Formation

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1 Introduction

As mentioned in the report which this is an appendix to, a stellar model must include a more detailed equation of acceleration than what is necessary for a galactic flyby. In this model, we include gravitational acceleration, a pressure gradient and the viscosity due to the other particles. A star collapses when a region of a molecular cloud becomes both dense and cold enough for the gravitational attraction to overcome the pressure gradient. However, once it starts collapsing, this gradient becomes larger again. This causes the acceleration to increase in the other direction, leading to the expansion of the molecular cloud. It keeps happening in turns and forever, unless something slows down the particles during one of the phases. This is why the viscosity is needed.

In this appendix, we will see how this model was implemented, then some results with different initial parameters for the simulations. Finally, we will take a look at what worked and what could be improved to model stellar formation as accurately as possible. Part of this appendix will be taken or inspired from the initial report for ease of understanding.

2 Implementation

The largest part of the model was implemented in the same way as for the galactic flyby, with a main file calling multiple functions. The first created an N-body cloud, the second calculated an initial acceleration and density map, then the position, velocity, density map and acceleration were updated using a leapfrog algorithm in the third. The last function plotted the updated positions and density map after each time steps.

The running time was only improved by the use of array maths, and masks instead of if-statements. Numba could not be used as functions were being called within loops. However, the code ran quite fast without this, with runtimes of at most a few hours for 2000 particles and a density resolution of 100×100 . Simulations with a factor 10 less particles and half the resolution ran in less than an hour.

2.1 Equation of motion

As mentioned in the [Introduction](#) and in the main report, the equation of motion was composed of three parts: the gravitational acceleration, the pressure gradient and the viscosity.

$$\frac{d\vec{v}}{dt} = \vec{g} - \nabla P - \nu \vec{v}$$

Where $\vec{g}$ is simply the acceleration obtained from Newton's law and the law of gravitational attraction:

$$\vec{F} = m\vec{a} = G \frac{m_j \cdot m_i}{r_{ji}^3}$$

It was implemented as such in the model:

$$\vec{a}_j = G \sum_{i \neq j} \frac{m_i}{r_{ji}^3} \vec{r}_{ji}.$$

Then the pressure gradient was also a function of the distance to all the other particles. However, the particles further away had less impact, and so this had to be reflected in the acceleration calculations. First, the equation of state was isothermal, so $P = \kappa \rho^2$. This being an N-body simulation, the true density could not be taken to simply be the mass per volume. So, we defined a smoothing kernel, W , which smoothed the point mass of each particle onto a larger surface. The value of this kernel depended on the distance to the centre of the point mass represented by $q = |\vec{r} - \vec{r}_j|/h$, h a constant, called the smoothing distance.

$$W_i(q, h) = C \begin{cases} (2 - q)^3 - 4(1 - q^3) & \text{if } 1 > q \geq 0, \\ (2 - q)^3 & \text{if } 2 > q \geq 1, \\ 0 & \text{if } q \geq 2. \end{cases}$$

C is another constant which also depends on h , $C = 5/(14\pi h^2)$.

As we saw above, $P = \kappa \rho^2$, so

$$\begin{aligned} \nabla P &= \kappa \nabla(\rho^2). \\ \Leftrightarrow \rho(\vec{r}_j) &= \sum_i m_i W_i(|\vec{r} - \vec{r}_i|, h) \\ \Leftrightarrow \nabla P_j &= -2\kappa \sum_i m_i \nabla_j W(|\vec{r}_j - \vec{r}_i|, h). \end{aligned}$$

And :

$$\nabla_i W_j(q, h) = -3 \cdot \frac{C}{h} \begin{cases} (2 - q)^2 - 4(1 - q^2) & \text{if } 1 > q \geq 0, \\ (2 - q)^2 & \text{if } 2 > q \geq 1, \\ 0 & \text{if } q \geq 2. \end{cases}$$

Finally, the viscosity is simply the velocity multiplied by a constant. The values of all the constants and initial parameters can be found in [Appendix A](#).

3 Results

A large set of simulations were run, with varying radius and mass, which in turn also varied the smoothing factor as it is directly dependent on the radius of the star. It also modified the minimum value the distance between particles could take within the equation for the gravitational acceleration as this was taken as $0.1R_{star}$. In total, 13 simulations were run, which are summarised in [Appendix A](#). The masses taken ranged from $0.1M_{\odot}$ to $M_{\odot}$, and the radii from $0.1R_{\odot}$ to $R_{\odot}$. In addition, one simulation was run with $R_{\odot}$ and $M_{\odot}$ but a smoothing factor of $0.1R_{star}$ instead of $0.02R_{star}$. As the simulations were sometimes paused, it is not possible to accurately estimate their runtime, but the maximum time recorded was of 7h, with a large break of unknown time taken during it.

A first point of interest is that for a same mass, stars of decreasing radius appeared larger in the plots (see fig. 1). These are simulations for stars of the mass of the Sun, and radii of $R_{\odot}$ (left), $0.7R_{\odot}$ (middle) and $0.3R_{\odot}$ (right). While it was run, the simulation of the same mass with a radius of $0.1R_{\odot}$ is not shown here because the particles were ejected within 50 time steps as they formed too close to each others.

4 Discussion

While this model accurately represents a cloud of particle falling into a sphere, there are a few elements missing to model the formation of a star. In particular, effects linked to changes in temperature. As we are using an isothermal equation of state, this model is entirely independent of heat, which is not an accurate representation. Jean's mass, which is the mass a molecular cloud needs to reach to collapse is dependent on both density and temperature. We are also not including radiation pressure, which would push particles away from each others. This model could represent a planet collapsing just as well as a star. Magneto-hydrodynamics are also missing. As stellar formation discs and young stars have very powerful magnetic fields, this could also influence their formation.

It is also a closed box model, no particles can enter, and none can escape due to another object. Stars form in groups with other stars and not alone, within molecular clouds. Not all particles present at $t = 0$ will fall into this star, and new particles could enter after the initial collapse due to some adjacent event.

This is still a reasonably accurate model to represent the way gravitational collapse happen, and the alternance of contractions and expansions which lead to stellar formation. It is however missing a few phenomena to represent a star and not any body forming from a group of particles.

The code can be found in [this repository](#), along with the L^AT_EX version of this report and the plots made during the simulations

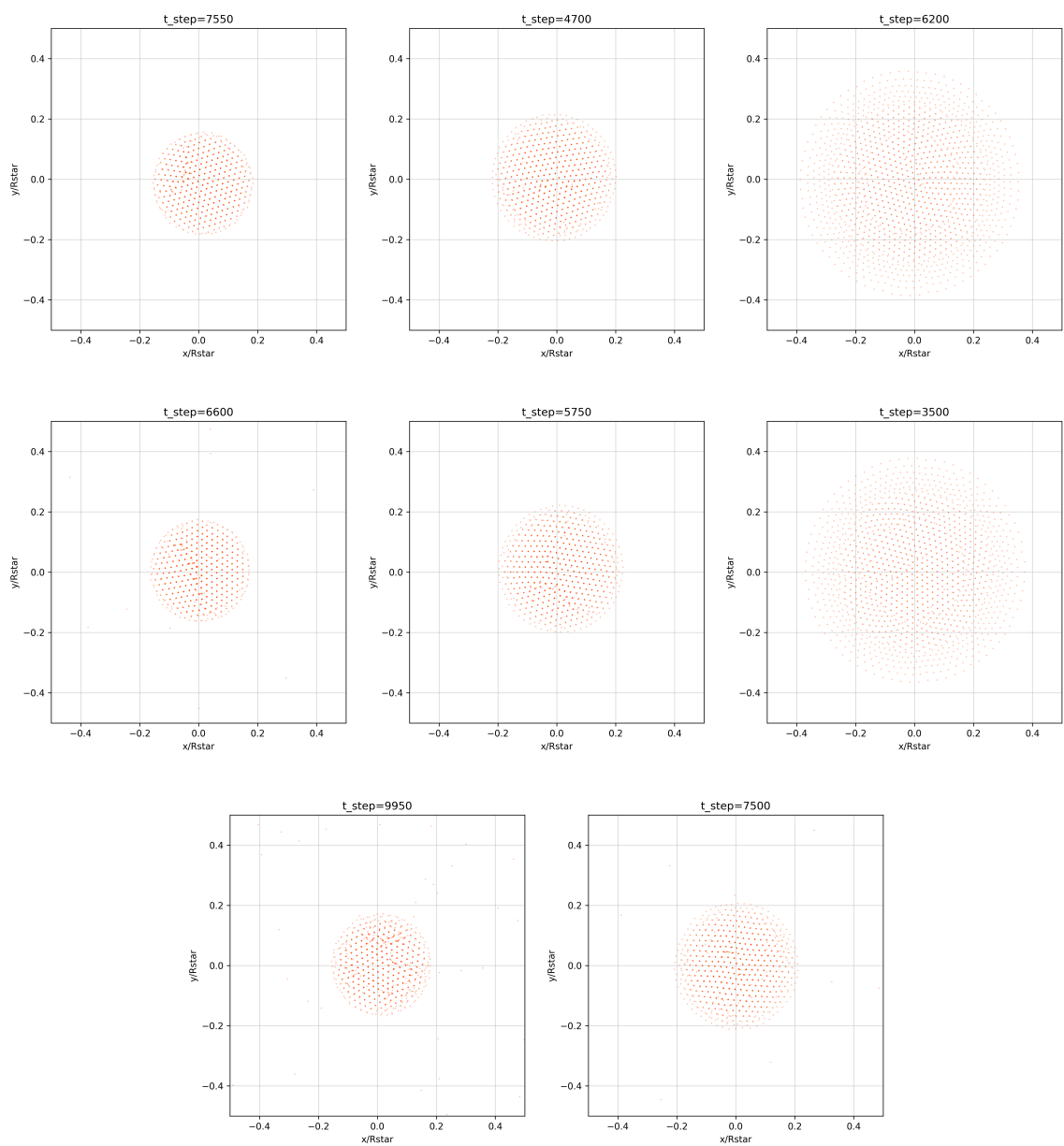


Figure 1

5 Appendices

5.1 Appendix A

Common to all simulations	
$M_{\odot}$	$1.99 \cdot 10^{30}$ Kg
$R_{\odot}$	$6.9 \cdot 10^{11}$ m
τ	$(\frac{R_{\odot}^3}{G \cdot M_g})^{0.5}$
dt	$0.00001 \cdot \tau$
n_{dim}	2
n_{steps}	4000
Simulations	
N_1	2000
$M_1 (M_{\odot})$	0.1, 0.5, 1
$R_1 (R_{\odot})$	0.1, 0.3, 0.7, 1
κ	10^{-2}
ν	$2 \cdot 10^{-3}$